

# EXPLICIT FORMULA FOR THE DISCRETE LAPLACE TRANSFORM OF THE MÖBIUS FUNCTION, RELATED SPECIAL FUNCTIONS, AND A CRITERION FOR THE RIEMANN HYPOTHESIS

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ABSTRACT. In this paper, we assume that all the zeros of the Riemann zeta function are simple. Under this assumption we give an explicit formula for the function  $\Phi(e^{-t}) = \sum_{n=1}^{\infty} \mu(n)e^{-nt}$ , as a function of the values of  $\zeta(s)$  and  $\zeta'(s)$  at the odd integers and as a function of the zeros of  $\zeta(s)$ . A structural feature distinguishes this formula from the classical explicit formula for the Mertens function: the poles of  $\Gamma(s)$  collide with the trivial zeros of  $\zeta(s)$ , producing double poles whose residues contain a logarithmic term. Using this formula, we give a criterion for the Riemann hypothesis: the bound  $O(x^{-1/2})$  on the transform implies the Riemann hypothesis unconditionally, while the converse direction requires additional hypotheses on the zeros. We also introduce special entire functions related to  $\zeta(s)$  and show that they admit absolutely convergent closed forms as Möbius-weighted series of Bessel functions of rotated argument.

## 1. THE MELLIN INTEGRAL REPRESENTATION

Here we give an explicit formula for the function

$$(1.1) \quad \Phi(x) = \sum_{n=1}^{\infty} \mu(n)x^n,$$

where  $\mu(n)$  is the Möbius function, under the assumption that the Riemann zeta function has only simple zeros in the critical strip.

We consider the identity (see [16, Chapter I]) given by

$$(1.2) \quad \frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} \quad (\operatorname{Re} s > 1).$$

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We observe that

$$(1.3) \quad \int_0^\infty e^{-nt} t^{s-1} dt = \frac{\Gamma(s)}{n^s}.$$

Since

$$\sum_{n=1}^\infty |\mu(n)| \int_0^\infty e^{-nt} t^{\sigma-1} dt \leq \Gamma(\sigma) \sum_{n=1}^\infty n^{-\sigma} = \Gamma(\sigma) \zeta(\sigma) < \infty$$

for every  $\sigma > 1$ , Tonelli's theorem permits the interchange of summation and integration, and we obtain that

$$(1.4) \quad \sum_{n=1}^\infty \mu(n) \int_0^\infty e^{-nt} t^{s-1} dt = \sum_{n=1}^\infty \mu(n) \frac{\Gamma(s)}{n^s} = \frac{\Gamma(s)}{\zeta(s)},$$

and on the other hand, by the same interchange,

$$(1.5) \quad \frac{\Gamma(s)}{\zeta(s)} = \int_0^\infty \left( \sum_{n=1}^\infty \mu(n) e^{-nt} \right) t^{s-1} dt = \int_0^\infty \Phi(e^{-t}) t^{s-1} dt.$$

Invoking the Mellin inversion theorem (see [5]) we obtain that, for any fixed  $\sigma > 1$  and every  $x > 0$ ,

$$(1.6) \quad \Phi(e^{-x}) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{\Gamma(s)}{\zeta(s)} x^{-s} ds.$$

The inversion is justified as follows. For fixed  $\sigma$  and  $|t| \rightarrow \infty$  we have the asymptotic estimate (see [7, Section 5.11] or [13, Section 1.4])

$$(1.7) \quad |\Gamma(\sigma + it)| \sim \sqrt{2\pi} e^{-\frac{1}{2}\pi|t|} |t|^{\sigma-\frac{1}{2}}.$$

Let us show that  $\zeta(\sigma + it)$  is bounded below on every fixed line  $\sigma > 1$ . Indeed, for  $\sigma > 1$  we have Euler's product formula ([16, Chapter I])

$$(1.8) \quad \zeta(s) = \prod_{p \in P} \frac{p^s}{p^s - 1},$$

and therefore

$$(1.9) \quad |\zeta(\sigma + it)| = \prod_{p \in P} \frac{|p^{\sigma+it}|}{|p^{\sigma+it} - 1|}.$$

We observe that  $|p^{\sigma+it}| = p^\sigma$  and that

$$(1.10) \quad |p^{\sigma+it} - 1| = (p^{2\sigma} - 2p^\sigma \cos(t \log p) + 1)^{\frac{1}{2}} \leq p^\sigma + 1.$$

Therefore

$$(1.11) \quad |\zeta(\sigma + it)| \geq \prod_{p \in P} \frac{p^\sigma}{p^\sigma + 1} = \prod_{p \in P} (1 + p^{-\sigma})^{-1} = \frac{\zeta(2\sigma)}{\zeta(\sigma)} =: \delta(\sigma) > 0,$$

where the constant  $\delta(\sigma)$  depends on  $\sigma$  (and tends to 0 as  $\sigma \rightarrow 1^+$ ; the bound is used only on fixed lines). Now let us estimate the integral in (1.6). By the triangle inequality ( $\sigma > 1$ ),

$$(1.12) \quad \left| \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\Gamma(\sigma + it)}{\zeta(\sigma + it)} x^{-\sigma - it} dt \right| \leq \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|\Gamma(\sigma + it)|}{|\zeta(\sigma + it)|} x^{-\sigma} dt \\ \leq C(\sigma) x^{-\sigma} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\pi|t|} |t|^{\sigma - \frac{1}{2}} dt$$

for a suitable constant  $C(\sigma)$ , by (1.7) and (1.11). Finally,

$$(1.13) \quad \int_{-\infty}^{\infty} e^{-\frac{1}{2}\pi|t|} |t|^{\sigma - \frac{1}{2}} dt = 2 \int_0^{\infty} e^{-\frac{1}{2}\pi t} t^{\sigma + \frac{1}{2} - 1} dt = \frac{2\Gamma(\sigma + \frac{1}{2})}{(\frac{1}{2}\pi)^{\sigma + \frac{1}{2}}}.$$

Thus the integral converges absolutely for every  $0 < x < \infty$ , and for every fixed  $\sigma > 1$  there exists  $M(\sigma)$  such that

$$(1.14) \quad |\Phi(e^{-x})| = \left| \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \frac{\Gamma(s)}{\zeta(s)} x^{-s} ds \right| \leq \frac{M(\sigma)}{x^{\sigma}}.$$

The hypotheses of the inversion theorem hold here with room to spare: the function  $t \mapsto \Phi(e^{-t})$  is continuous and of bounded variation on every compact subinterval of  $(0, \infty)$ , being a locally uniformly convergent series of smooth functions whose termwise derivative also converges locally uniformly (Weierstrass test against  $\sum_n n e^{-nt_0}$  for  $t \geq t_0 > 0$ ); and its Mellin transform converges absolutely on every line  $\sigma > 1$ , since

$$\int_0^{\infty} |\Phi(e^{-t})| t^{\sigma - 1} dt \leq \int_0^{\infty} \sum_{n \geq 1} e^{-nt} t^{\sigma - 1} dt = \Gamma(\sigma) \zeta(\sigma) < \infty.$$

(We remark in passing that the elementary bound

$$|\Phi(e^{-x})| \leq \sum_{n \geq 1} e^{-nx} = (e^x - 1)^{-1} \leq x^{-1}$$

holds trivially for all  $x > 0$ ; the content of (1.6) is the identity, not the bound.)

**1.1. Statement of the main results.** Vast numerical calculations support the conjecture that all the nontrivial zeros of the Riemann zeta function are simple; in particular, all zeros in the range in which the Riemann hypothesis has been verified, up to height about  $3 \cdot 10^{12}$ , are simple [14]. We refer to this assumption as Hypothesis (S). Throughout,  $\rho = \beta_{\rho} + i\gamma$  denotes a nontrivial zero of  $\zeta(s)$ , so that  $\gamma = \text{Im } \rho$ , and sums over  $\rho$  are sums over the nontrivial zeros. Because the series over the zeros is not known to converge absolutely,

we adopt the standard convention of symmetric partial sums: there is a sequence of heights  $T_\nu \rightarrow \infty$ , provided by a classical selection lemma of Titchmarsh recalled as Lemma 5.2 below and equal to no ordinate (an *ordinate* is the imaginary part  $\gamma$  of a nontrivial zero; thus no zero lies on the lines  $t = \pm T_\nu$ ), and we define

$$(1.15) \quad \sum_{\rho} a_{\rho} := \lim_{\nu \rightarrow \infty} \sum_{\rho: |\gamma| < T_\nu} a_{\rho},$$

exactly as in the explicit formula for the Mertens function [16, Theorem 14.27]. Under the mild Hypothesis (H) introduced in Section 7 the series converges absolutely and the limit in (1.15) is superfluous.

We shall prove the following results. Define the entire functions

$$(1.16) \quad \begin{aligned} \kappa(x) &= \sum_{m=0}^{\infty} \frac{(-1)^m (2\pi x)^{2m+1}}{\zeta(2m+2) [(2m+1)!]^2}, \\ \beta(x) &= \sum_{m=1}^{\infty} \frac{(-1)^m (2\pi x)^{2m}}{[(2m)!]^2 \zeta(2m+1)}, \\ \lambda(x) &= \sum_{m=1}^{\infty} \frac{(-1)^m \zeta'(2m+1) (2\pi x)^{2m}}{[(2m)!]^2 \zeta(2m+1)^2}, \\ v(x) &= \sum_{m=1}^{\infty} \frac{(-1)^m \Gamma'(2m+1) (2\pi x)^{2m}}{[(2m)!]^3 \zeta(2m+1)}. \end{aligned}$$

**Theorem 1.1** (Explicit formula). *Assume Hypothesis (S). Then for every  $x > 0$ , with the convention (1.15),*

$$(1.18) \quad \sum_{n=1}^{\infty} \mu(n) e^{-nx} = \sum_{\rho} \frac{\Gamma(\rho) x^{-\rho}}{\zeta'(\rho)} + \pi \kappa(x) + 2\lambda(x) + 4v(x) - 2\beta(x) \log(2\pi x) - 2.$$

**Theorem 1.2** (Möbius–Bessel expansions). *For every  $x > 0$  the following series converge absolutely:*

$$(1.19) \quad \sum_{k=1}^{\infty} \frac{\mu(k)}{k} \left( J_0 \left( 2e^{-i\pi/4} \sqrt{\frac{2\pi x}{k}} \right) - 1 \right) = \beta(x) + i \kappa(x),$$

$$(1.20) \quad \sum_{k=1}^{\infty} \frac{\mu(k) \log k}{k} \operatorname{Re} \left( J_0 \left( 2e^{-i\pi/4} \sqrt{\frac{2\pi x}{k}} \right) - 1 \right) = \lambda(x),$$

where  $J_0$  is the Bessel function of the first kind of order zero. Moreover  $\kappa, \beta, \lambda, v$  are entire functions.

**Theorem 1.3** (A criterion for the Riemann hypothesis).

(a) *If*

$$(1.21) \quad \sum_{n=1}^{\infty} \mu(n) e^{-nx} = O(x^{-1/2}) \quad \text{as } x \rightarrow 0^+,$$

*then the Riemann hypothesis is true. No hypothesis on the zeros is required.*

(b) *Conversely, if the Riemann hypothesis, Hypothesis (S), and Hypothesis (H) of Section 7 hold, then (1.21) holds.*

Let us describe what is new here. The method, Mellin inversion followed by a shift of the contour, is the classical one of Hardy and Littlewood [10], who studied the closely related Gaussian-kernel sum

$$\sum_{n=1}^{\infty} \frac{\mu(n)}{n} e^{-x/n^2}$$

arising from an identity on page 312 of Ramanujan's second notebook (see Berndt [4, pp. 468–469]); the criteria of Riesz [15] and of Hardy–Littlewood belong to the same family, which remains active [1, 8]. For the Mertens function itself the corresponding explicit formula is Theorem 14.27 of Titchmarsh [16], proved unconditionally by Bartz [3]; see also [12]. In all of those formulas the trivial zeros contribute through *simple* poles. The Laplace kernel treated here is distinguished by the collision of the poles of  $\Gamma(s)$  with the trivial zeros of  $\zeta(s)$ : the integrand of (1.6) has *double* poles at the negative even integers, and their residues, computed in Section 4, contain the term  $-2\beta(x) \log(2\pi x)$  together with the digamma weights  $\psi(2m+1)$  and the weights  $\zeta'(2m+1)/\zeta(2m+1)$  visible in (1.17). This logarithmic structure has no analogue in (1.22) below and appears to be new. The second novelty is Theorem 1.2: the special functions generated by the trivial zeros are Möbius-weighted Bessel series of rotated argument, in complete analogy with the Bessel-type functions appearing in the Riesz and Hardy–Littlewood criteria. The third is the asymmetry in Theorem 1.3: direction (a) is unconditional.

To summarize, the contributions of this paper are threefold:

- (1) the exact explicit formula (1.18), in which the collision of the poles of  $\Gamma(s)$  with the trivial zeros of  $\zeta(s)$  produces double poles and the logarithmic term  $-2\beta(x) \log(2\pi x)$ , a structure absent from all Mertens-type explicit formulas;
- (2) the closed forms of Theorem 1.2, expressing the trivial-zero contributions as absolutely convergent Möbius-weighted Bessel

series of rotated argument, in complete analogy with the Bessel-type functions of the Riesz and Hardy–Littlewood criteria;

- (3) the criterion of Theorem 1.3, whose direction (a) is unconditional, together with a fully detailed contour proof whose key ingredient, the uniform lower bound of Lemma 5.1, replaces the fixed-abscissa Stirling asymptotic (1.7), which fails on the moving line  $\operatorname{Re} s = \frac{1}{2} - 2N$ .

For comparison we record the Mertens-function formula: under RH and Hypothesis (S),

$$(1.22) \quad \sum'_{n \leq x} \mu(n) = \lim_{\nu \rightarrow \infty} \sum_{|\gamma| < T_\nu} \frac{x^\rho}{\rho \zeta'(\rho)} - 2 + \sum_{n=1}^{\infty} \frac{(-1)^{n-1} (2\pi/x)^{2n}}{(2n)! n \zeta(2n+1)},$$

where the last series comes from simple poles at the trivial zeros; the constant  $-2$  has the same origin, the residue at  $s = 0$ , in both (1.18) and (1.22).

## 2. CALCULATION OF THE RESIDUES IN THE CRITICAL STRIP

We need Hypothesis (S) here since it allows us to calculate explicitly the residues of the function  $\Gamma(s)x^{-s}/\zeta(s)$  at the zeros of zeta. Let  $\rho$  denote a nontrivial zero of zeta. Using the assumption that this zero is simple and that  $\Gamma(s)$  has no zeros in the complex plane, we obtain by L'Hôpital's rule that

$$(2.1) \quad \operatorname{Res}_{s=\rho} \frac{\Gamma(s)x^{-s}}{\zeta(s)} = \frac{\Gamma(\rho)x^{-\rho}}{\zeta'(\rho)}.$$

Next,  $\Gamma(s)$  has a simple pole at  $s = 0$ , and  $\zeta(0) = -\frac{1}{2}$ . Let us first calculate the residues of  $\Gamma(s)$  at the nonpositive integers. We use the reflection identity ([7, Section 5.5])

$$(2.2) \quad \Gamma(s) \Gamma(1-s) = \frac{\pi}{\sin(\pi s)}.$$

Since  $\Gamma(s)$  has no zeros, and since  $\sin(\pi s) = (-1)^n \sin(\pi(s+n))$  near  $s = -n$ ,

$$(2.3) \quad \lim_{s \rightarrow -n} \frac{(s+n)\pi}{\sin(\pi s)} = \frac{1}{\cos(\pi n)} = (-1)^n,$$

and we obtain that

$$(2.4) \quad \operatorname{Res}_{s=-n} \Gamma(s) = \lim_{s \rightarrow -n} \frac{(s+n)\pi}{\sin(\pi s) \Gamma(1-s)} = \frac{(-1)^n}{\Gamma(1+n)} = \frac{(-1)^n}{n!}.$$

More precisely, we shall use the Laurent expansion at  $s = -n$ :

$$(2.5) \quad \Gamma(s) = \frac{(-1)^n}{n!} \left( \frac{1}{s+n} + \psi(n+1) + O(s+n) \right) \quad (s \rightarrow -n),$$

where  $\psi = \Gamma'/\Gamma$  is the digamma function; the constant term follows from (2.2) and the expansion of  $\pi/\sin(\pi s)$  or, alternatively, from  $\Gamma(s) = \Gamma(s+n+1)/(s(s+1)\cdots(s+n))$ ; see [13, Section 1.1].

By (2.4),  $\text{Res}_{s=0} \Gamma(s) = 1$ , and therefore

$$(2.6) \quad \text{Res}_{s=0} \frac{\Gamma(s)x^{-s}}{\zeta(s)} = \lim_{s \rightarrow 0} \frac{s \Gamma(s) x^{-s}}{\zeta(s)} = \frac{1}{\zeta(0)} = -2.$$

In this section we have shown, under Hypothesis (S), that the sum of the residues of  $\Gamma(s)x^{-s}/\zeta(s)$  at  $s = 0$  and at the zeros in the critical strip is

$$(2.7) \quad -2 + \sum_{\rho} \frac{\Gamma(\rho) x^{-\rho}}{\zeta'(\rho)},$$

with the summation convention (1.15).

### 3. CALCULATION OF THE RESIDUES AT THE ODD NEGATIVE INTEGERS

First we recall the Bernoulli numbers, which are determined by the power series expansion

$$(3.1) \quad \frac{t}{e^t - 1} = \sum_{n=0}^{\infty} \frac{B_n t^n}{n!},$$

so that  $B_{2n+1} = 0$  for  $n \geq 1$ . Using the classical contour integral representation of  $\zeta(s)$  and the residue theorem (see [16, Chapter II] or [2]) one obtains Euler's evaluation

$$(3.2) \quad \zeta(2m) = (-1)^{m+1} \frac{(2\pi)^{2m} B_{2m}}{2 (2m)!},$$

or equivalently

$$(3.3) \quad B_{2m+2} = (-1)^m \frac{2 (2m+2)! \zeta(2m+2)}{(2\pi)^{2m+2}}.$$

The analytic continuation of  $\zeta(s)$  to  $\mathbb{C} \setminus \{1\}$  is given by the celebrated functional equation (see [16, Chapter II])

$$(3.4) \quad \zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{1}{2}\pi s\right) \Gamma(1-s) \zeta(1-s).$$

Equation (3.4) implies that  $\zeta(s)$  has simple zeros at the points  $s = -2, -4, -6, \dots$ , the trivial zeros, and it also allows us to calculate the values of  $\zeta(s)$  at the odd negative integers. Namely,

$$(3.5) \quad \zeta(-(2m+1)) = 2^{-(2m+1)} \pi^{-2m-2} \sin\left(-\frac{1}{2}\pi + \pi m\right) \Gamma(2m+2) \zeta(2m+2),$$

Here  $\sin\left(\pi m - \frac{\pi}{2}\right) = -\cos(\pi m) = (-1)^{m+1}$  and  $\Gamma(2m+2) = (2m+1)!$ , so that

$$(3.6) \quad \zeta(-(2m+1)) = (-1)^{m+1} \frac{(2m+1)! \zeta(2m+2)}{2^{2m+1} \pi^{2m+2}},$$

and substituting (3.3) gives

$$(3.7) \quad \zeta(-(2m+1)) = -\frac{B_{2m+2}}{2m+2} \quad (m \geq 0).$$

Now, since the pole of  $\Gamma(s)$  at  $-(2m+1)$  is simple with residue  $-1/(2m+1)!$  by (2.4), while  $\zeta(s)$  is regular and nonzero there by (3.7), we calculate

$$(3.8) \quad \begin{aligned} \operatorname{Res}_{s=-(2m+1)} \frac{\Gamma(s)x^{-s}}{\zeta(s)} &= \lim_{s \rightarrow -(2m+1)} (s+2m+1) \frac{\Gamma(s)x^{-s}}{\zeta(s)} \\ &= \frac{(-1)^{2m+1}}{(2m+1)!} \cdot \frac{x^{2m+1}}{\zeta(-(2m+1))} = -\frac{x^{2m+1}}{(2m+1)!} \cdot \frac{2m+2}{B_{2m+2}} \\ &= \frac{(2m+2)x^{2m+1}}{(2m+1)! B_{2m+2}}. \end{aligned}$$

Thus we have

$$(3.9) \quad \sum_{m=0}^{\infty} \operatorname{Res}_{s=-(2m+1)} \frac{\Gamma(s)x^{-s}}{\zeta(s)} = \sum_{m=0}^{\infty} \frac{(2m+2)x^{2m+1}}{(2m+1)! B_{2m+2}},$$

and using (3.3) we obtain

$$(3.10) \quad \sum_{m=0}^{\infty} \operatorname{Res}_{s=-(2m+1)} \frac{\Gamma(s)x^{-s}}{\zeta(s)} = \pi \sum_{m=0}^{\infty} \frac{(-1)^m (2\pi x)^{2m+1}}{\zeta(2m+2) [(2m+1)!]^2}.$$

Here we recognize the function  $\kappa(x)$  of (1.16), which admits the two equivalent representations

$$(3.11) \quad \kappa(x) = \sum_{m=0}^{\infty} \frac{(-1)^m (2\pi x)^{2m+1}}{\zeta(2m+2) [(2m+1)!]^2} = \sum_{m=0}^{\infty} \frac{(2m+2)x^{2m+1}}{\pi (2m+1)! B_{2m+2}}.$$

This function seems to be very interesting in its own right; its closed form as a Möbius–Bessel series is given in Theorem 1.2 and proved in



Section 6. We conclude this section with the main result:

$$(3.12) \quad \sum_{m=0}^{\infty} \operatorname{Res}_{s=-(2m+1)} \frac{\Gamma(s)x^{-s}}{\zeta(s)} = \pi \kappa(x).$$

#### 4. CALCULATION OF THE RESIDUES AT THE EVEN NEGATIVE INTEGERS

At the negative even integers  $\zeta(s)$  has simple zeros and  $\Gamma(s)$  has simple poles, so the function  $\Gamma(s)x^{-s}/\zeta(s)$  has poles of order 2 at the negative even integers. Therefore

$$(4.1) \quad \operatorname{Res}_{s=-2m} \frac{\Gamma(s)x^{-s}}{\zeta(s)} = \lim_{s \rightarrow -2m} \frac{d}{ds} \left( \frac{(s+2m)^2 \Gamma(s) x^{-s}}{\zeta(s)} \right).$$

Now, using (2.2) and (3.4), we can write

$$(4.2) \quad \frac{(s+2m)^2 \Gamma(s) x^{-s}}{\zeta(s)} = \frac{(s+2m)^2 x^{-s} \pi}{2^s \pi^{s-1} \Gamma(1-s)^2 \zeta(1-s) \sin(\frac{1}{2}\pi s) \sin(\pi s)}.$$

Let us denote

$$(4.3) \quad f(s) = \frac{\pi x^{-s}}{2^s \pi^{s-1} \Gamma(1-s)^2 \zeta(1-s)} = \frac{\pi^2}{(2\pi x)^s \Gamma(1-s)^2 \zeta(1-s)}.$$

Then for  $\sigma < 0$  the function  $f(s)$  is analytic with no zeros and no poles, and we have the limit

$$(4.4) \quad \begin{aligned} \lim_{s \rightarrow -2m} \frac{d}{ds} \left( f(s) \frac{(s+2m)^2}{\sin(\frac{1}{2}\pi s) \sin(\pi s)} \right) \\ = \lim_{s \rightarrow -2m} \left( f'(s) \frac{(s+2m)^2}{\sin(\frac{1}{2}\pi s) \sin(\pi s)} + f(s) \frac{d}{ds} \frac{(s+2m)^2}{\sin(\frac{1}{2}\pi s) \sin(\pi s)} \right). \end{aligned}$$

As is easily seen,

$$(4.5) \quad \lim_{s \rightarrow -2m} \frac{s+2m}{\sin(\pi s)} = \frac{1}{\pi \cos(2\pi m)} = \frac{1}{\pi},$$

and

$$(4.6) \quad \lim_{s \rightarrow -2m} \frac{s+2m}{\sin(\frac{1}{2}\pi s)} = \frac{1}{\frac{1}{2}\pi \cos(\pi m)} = (-1)^m \frac{2}{\pi},$$

so that, multiplying (4.5) and (4.6),

$$(4.7) \quad \lim_{s \rightarrow -2m} \frac{(s+2m)^2}{\sin(\frac{1}{2}\pi s) \sin(\pi s)} = \frac{2(-1)^m}{\pi^2}.$$

Next we differentiate  $f$ . Taking the logarithmic derivative of (4.3),

$$(4.8) \quad \log f(s) = 2 \log \pi - s \log(2\pi x) - 2 \log \Gamma(1-s) - \log \zeta(1-s),$$

and therefore

$$(4.9) \quad f'(s) = f(s) \left( \frac{\zeta'(1-s)}{\zeta(1-s)} + \frac{2\Gamma'(1-s)}{\Gamma(1-s)} - \log(2\pi x) \right),$$

since

$$\begin{aligned} \frac{d}{ds}(-s \log(2\pi x)) &= -\log(2\pi x), \\ \frac{d}{ds}(-2 \log \Gamma(1-s)) &= 2\psi(1-s) \end{aligned}$$

and

$$\frac{d}{ds}(-\log \zeta(1-s)) = \zeta'(1-s)/\zeta(1-s).$$

Substituting  $s = -2m$  we obtain

$$(4.10) \quad f'(-2m) = \frac{\pi^2 (2\pi x)^{2m}}{[(2m)!]^2 \zeta(2m+1)} \times \left( \frac{\zeta'(2m+1)}{\zeta(2m+1)} + \frac{2\Gamma'(2m+1)}{(2m)!} - \log(2\pi x) \right).$$

Here we substituted  $s = -2m$ , so that  $1-s = 2m+1$ ,  $\Gamma(2m+1) = (2m)!$ ,  $\zeta(1-s) = \zeta(2m+1)$  and  $(2\pi x)^{-s} = (2\pi x)^{2m}$ . For the second term in (4.4) we claim that

$$(4.11) \quad \lim_{s \rightarrow -2m} \frac{d}{ds} \frac{(s+2m)^2}{\sin(\frac{1}{2}\pi s) \sin(\pi s)} = 0.$$

Indeed, writing  $w = s + 2m$ , both  $\sin(\pi s) = \sin(\pi w)$  and  $\sin(\frac{1}{2}\pi s) = (-1)^m \sin(\frac{1}{2}\pi w)$  are odd functions of  $w$ , so their product is an even function of  $w$  with a double zero at  $w = 0$ ; hence there are coefficients  $c_0 \neq 0, c_2, c_4, \dots$  such that near  $w = 0$

$$(4.12) \quad \frac{w^2}{\sin(\frac{1}{2}\pi s) \sin(\pi s)} = c_0 + c_2 w^2 + c_4 w^4 + \dots,$$

an even analytic function of  $w$  whose derivative vanishes at  $w = 0$ . This proves (4.11).

Combining (4.4), (4.7), (4.10) and (4.11), we obtain the residue at the double pole:

$$(4.13) \quad \operatorname{Res}_{s=-2m} \frac{\Gamma(s)x^{-s}}{\zeta(s)} = \frac{2(-1)^m (2\pi x)^{2m}}{[(2m)!]^2 \zeta(2m+1)} \times \left( \frac{\zeta'(2m+1)}{\zeta(2m+1)} + \frac{2\Gamma'(2m+1)}{(2m)!} - \log(2\pi x) \right).$$

An instructive consistency check, the direct Laurent computation of the same residue through the expansions of  $\Gamma$  and of  $1/\zeta$  at

$s = -2m$  together with the functional-equation evaluations of  $\zeta'(-2m)$  and  $\zeta''(-2m)$ , is carried out in full in Appendices C–G and reproduces (4.13) exactly.

We now recall the functions  $\lambda(x)$  and  $v(x)$  of (1.17) and the function  $\beta(x)$  of (1.16). Summing (4.13) over  $m \geq 1$  and separating the three terms in the bracket, we conclude this section with the final result:

$$(4.14) \quad \sum_{m=1}^{\infty} \operatorname{Res}_{s=-2m} \frac{\Gamma(s)x^{-s}}{\zeta(s)} = 2\lambda(x) + 4v(x) - 2\beta(x) \log(2\pi x).$$

Finally, combining (2.7), (3.12) and (4.14): if all the nontrivial zeros of the Riemann zeta function are simple, then the sum of the residues of  $\Gamma(s)x^{-s}/\zeta(s)$  over all poles with  $\operatorname{Re} s < \frac{3}{2}$  is

$$(4.15) \quad \sum_{\rho} \frac{\Gamma(\rho) x^{-\rho}}{\zeta'(\rho)} + \pi\kappa(x) + 2\lambda(x) + 4v(x) - 2\beta(x) \log(2\pi x) - 2.$$

## 5. THE INTEGRAL

In this section we prove Theorem 1.1 by moving the line of integration in (1.6) to the left and showing that the contribution of the auxiliary contour tends to zero. We first collect the identities and estimates that will be needed.

**5.1. The reflected representation.** First let us observe that, by (2.2),

$$(5.1) \quad \Gamma(s) = \frac{\pi}{\sin(\pi s) \Gamma(1-s)},$$

and by the functional equation (3.4),

$$(5.2) \quad \frac{1}{\zeta(s)} = \frac{1}{2^s \pi^{s-1} \sin(\frac{1}{2}\pi s) \Gamma(1-s) \zeta(1-s)}.$$

This implies, for the integrand  $F(s) = \Gamma(s)x^{-s}/\zeta(s)$ ,

$$(5.3) \quad F(s) = \frac{\pi^2 (2\pi x)^{-s}}{\sin(\pi s) \sin(\frac{1}{2}\pi s) \Gamma(1-s)^2 \zeta(1-s)}.$$

The representation (5.3) will be used on the part of the contour with  $\sigma \leq 0$ , where it converts all the quantities to be estimated into quantities evaluated in the half-plane  $\operatorname{Re}(1-s) \geq 1$ .

Also, writing  $\sin(\sigma + it) = \sin(\sigma) \cosh(it) + \sin(it) \cos(\sigma)$  and using the Euler formulas

$$(5.4) \quad \begin{aligned} \sin(it) &= \frac{e^{i(it)} - e^{-i(it)}}{2i} = i \sinh(t), \\ \cos(it) &= \frac{e^{i(it)} + e^{-i(it)}}{2} = \cosh(t), \end{aligned}$$

we obtain

$$(5.5) \quad \sin(\sigma + it) = \sin(\sigma) \cosh(t) + i \cos(\sigma) \sinh(t),$$

$$(5.6) \quad |\sin(\sigma + it)|^2 = \sin^2(\sigma) + \sinh^2(t).$$

In particular, on the vertical line  $\sigma = \frac{1}{2} - 2N$  ( $N \in \mathbb{N}$ ) we get the exact values

$$(5.7) \quad \left| \sin\left(\pi\left(\frac{1}{2} - 2N + it\right)\right) \right| = \cosh(\pi t) \geq \frac{1}{2} e^{\pi|t|},$$

$$(5.8) \quad \left| \sin\left(\frac{\pi}{2}\left(\frac{1}{2} - 2N + it\right)\right) \right| = \sqrt{\frac{1}{2} \cosh(\pi t)} \geq \frac{1}{\sqrt{2}},$$

(complete derivations of (5.7) and (5.8), including the exact evaluation of the shifted angles, are written out in Appendix A), and on horizontal lines  $t = \pm T$  with  $T \geq 1$ ,

$$(5.9) \quad \left| \sin\left(\pi(\sigma \pm iT)\right) \right| \geq \sinh(\pi T) \geq \frac{1}{4} e^{\pi T},$$

$$(5.10) \quad \left| \sin\left(\frac{\pi}{2}(\sigma \pm iT)\right) \right| \geq \sinh\left(\frac{\pi T}{2}\right) \geq \frac{1}{4} e^{\pi T/2}.$$

Again the four estimates are rederived step by step in Appendix A.

**5.2. Two lemmas.** The first lemma replaces the asymptotic estimate (1.7), which is valid for fixed  $\sigma$  only, by a lower bound valid uniformly in  $\sigma$ ; this uniformity is exactly what is needed on the moving line  $\sigma = \frac{1}{2} - 2N$ .

**Lemma 5.1** (Uniform lower bound for  $\Gamma$  on vertical lines). *For every  $\sigma > 0$  and every real  $t$ ,*

$$(5.11) \quad |\Gamma(\sigma + it)| \geq \Gamma(\sigma) \left(1 + \frac{t^2}{\sigma^2}\right)^{-1/2} e^{-\pi|t|/2}.$$

*Proof.* From the Weierstrass product

$$\frac{1}{\Gamma(z)} = z e^{\gamma_e z} \prod_{n \geq 1} \left(1 + \frac{z}{n}\right) e^{-z/n},$$

where  $\gamma_e$  is Euler's constant (see [17, Chapter XII]), applied at  $z = \sigma + it$  and at  $z = \sigma$ , and using  $|\sigma + it + n|^2 = (\sigma + n)^2 + t^2$ , one

obtains the exact identity

$$(5.12) \quad \frac{|\Gamma(\sigma + it)|^2}{\Gamma(\sigma)^2} = \prod_{n=0}^{\infty} \left(1 + \frac{t^2}{(n + \sigma)^2}\right)^{-1}.$$

Since the function  $u \mapsto \log(1 + t^2/u^2)$  is positive and decreasing on  $(0, \infty)$ , comparison of the sum with the integral gives

$$(5.13) \quad \sum_{n=0}^{\infty} \log \left(1 + \frac{t^2}{(n + \sigma)^2}\right) \leq \log \left(1 + \frac{t^2}{\sigma^2}\right) + \int_0^{\infty} \log \left(1 + \frac{t^2}{u^2}\right) du.$$

Substituting  $u = |t|v$  and integrating by parts,

$$\int_0^{\infty} \log(1 + v^{-2}) dv = [v \log(1 + v^{-2})]_0^{\infty} + \int_0^{\infty} \frac{2}{1+v^2} dv = 0 + \pi$$

, so the last integral equals  $\pi |t|$ . Exponentiating (5.13) and inserting it into (5.12) gives (5.11). Every step, including the derivation of the exact identity (5.12) from the Weierstrass product and the boundary term in the integration by parts, is written out in Appendix H.  $\square$

The second lemma provides horizontal lines that avoid the zeros of  $\zeta(s)$  quantitatively; it is classical.

**Lemma 5.2** (Ordinate selection). *There are an absolute constant  $A_0 > 0$  and heights  $T_\nu \in [\nu, \nu + 1]$ ,  $\nu = 2, 3, 4, \dots$ , such that*

$$(5.14) \quad \frac{1}{|\zeta(\sigma \pm iT_\nu)|} \leq T_\nu^{A_0} \quad (-1 \leq \sigma \leq 2).$$

*In particular  $\zeta$  has no zero on the two lines  $t = \pm T_\nu$ , so no singularity of  $F$  lies on them.*

*Remark 5.3* (Source). Lemma 5.2 is Theorem 9.7 of Titchmarsh [16, Theorem 9.7], and we do not reprove it; the bound holds uniformly in  $-1 \leq \sigma \leq 2$ , and for both signs of the imaginary part because  $\zeta(\bar{s}) = \overline{\zeta(s)}$ . The same selection device underlies the unconditional explicit formulas for the Mertens function of Bartz [3] and their descendants [12]; for the closely related classical selection of ordinates along which  $\zeta'(s)/\zeta(s) \ll \log^2 T$  on the same strip, see [6, Section 17]. The final assertion of the lemma is immediate: the bound (5.14) makes  $1/\zeta$  finite at every point of the two lines, so  $\zeta$  cannot vanish there.

*Remark 5.4* (Margin audit). The polynomial factor in (5.14) cannot cause divergence anywhere. It is used in exactly one estimate, the near horizontal segments in (5.22), where it meets the Stirling factor  $e^{-\pi T_N/2}$  of (5.21). On the right edge, on the far horizontal segments, and on the

left edge, the integrand is controlled through (1.11) and the reflected representation (5.3), in which the argument of  $\zeta$  has real part at least 2, and no ordinate bound enters.

**5.3. Proof of Theorem 1.1.** Fix  $x > 0$  and set  $\sigma_0 = \frac{3}{2}$ . For  $N \geq 2$  let  $T_N$  be as in Lemma 5.2 and let  $R_N$  denote the positively oriented rectangle with vertices

$$(5.15) \quad \sigma_0 \pm iT_N, \quad \left(\frac{1}{2} - 2N\right) \pm iT_N.$$

The left abscissa  $\frac{1}{2} - 2N$  lies strictly between the poles of  $F$  at  $-2N$  and at  $-(2N - 1)$ , and by the choice of  $T_N$  no singularity of  $F$  lies on the boundary. Inside  $R_N$  the integrand  $F$  has the following poles and no others: the nontrivial zeros  $\rho$  with  $|\gamma| < T_N$ ; the point  $s = 0$ ; the odd negative integers  $-(2m + 1)$  with  $0 \leq m \leq N - 1$ ; and the even negative integers  $-2m$  with  $1 \leq m \leq N - 1$ . (At  $s = 1$  the pole of  $\zeta(s)$  makes  $F$  vanish, so  $s = 1$  is a regular point.) The contour, its four segments  $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3, \mathcal{C}_4$ , its orientation, and the singularity structure of  $F$  are displayed in Figure 1. By the residue theorem and Sections 2, 3 and 4,

$$(5.16) \quad \frac{1}{2\pi i} \oint_{R_N} F(s) ds = \sum_{|\gamma| < T_N} \frac{\Gamma(\rho)x^{-\rho}}{\zeta'(\rho)} - 2 + \sum_{m=0}^{N-1} \operatorname{Res}_{-(2m+1)} F + \sum_{m=1}^{N-1} \operatorname{Res}_{-2m} F.$$

We now show that, as  $N \rightarrow \infty$ , the right edge  $\mathcal{C}_1$  of  $R_N$  produces the integral (1.6) while the remaining three segments  $\mathcal{C}_2, \mathcal{C}_3$  and  $\mathcal{C}_4$  tend to zero.

*Segment  $\mathcal{C}_1$ : the right edge (blue in Figure 1).* We work on the line  $\sigma = \sigma_0$ . There the integrand is absolutely integrable by (1.7) and (1.11), and the truncated integral differs from the full one by the two tails over  $|t| \geq T_N$ , each bounded in closed form via the upper incomplete Gamma function, defined for  $z > 0$  by ([7, Section 8.2])

$$\Gamma(a, z) = \int_z^\infty e^{-u} u^{a-1} du,$$

as follows:

$$\begin{aligned} C(\sigma_0) x^{-\sigma_0} \int_{T_N}^\infty e^{-\pi t/2} t^{\sigma_0 - \frac{1}{2}} dt \\ = C(\sigma_0) x^{-\sigma_0} \left(\frac{2}{\pi}\right)^{\sigma_0 + \frac{1}{2}} \Gamma\left(\sigma_0 + \frac{1}{2}, \frac{\pi}{2} T_N\right) \longrightarrow 0 \quad (N \rightarrow \infty), \end{aligned}$$

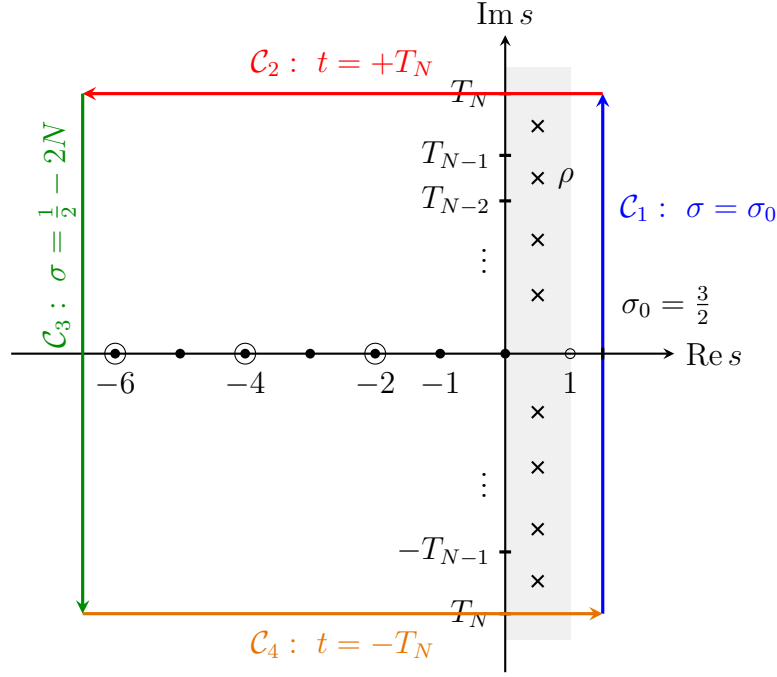


FIGURE 1. The positively oriented contour  $R_N = \mathcal{C}_1 \cup \mathcal{C}_2 \cup \mathcal{C}_3 \cup \mathcal{C}_4$  of (5.15): right edge  $\mathcal{C}_1$  (blue) on  $\sigma = \sigma_0$ , top edge  $\mathcal{C}_2$  (red) at  $t = T_N$ , left edge  $\mathcal{C}_3$  (green) on  $\sigma = \frac{1}{2} - 2N$ , bottom edge  $\mathcal{C}_4$  (orange) at  $t = -T_N$ ; the heights  $\pm T_\nu$  of Lemma 5.2 are marked on the imaginary axis. Crosses: nontrivial zeros of  $\zeta$ , drawn on the critical line (shaded). Dots: poles of  $\Gamma$ . Circled dots: collisions of the poles of  $\Gamma$  with the trivial zeros of  $\zeta$ , the double poles of  $F$ . Open circle: the regular point  $s = 1$ .

so

$$(5.17) \quad \frac{1}{2\pi i} \int_{\sigma_0 - iT_N}^{\sigma_0 + iT_N} F(s) ds \longrightarrow \frac{1}{2\pi i} \int_{\sigma_0 - i\infty}^{\sigma_0 + i\infty} F(s) ds = \Phi(e^{-x}) \quad (N \rightarrow \infty),$$

by (1.6).

*Segment  $\mathcal{C}_3$ : the left edge (green in Figure 1).* We work on the line  $\sigma = \frac{1}{2} - 2N$ : write  $s = \frac{1}{2} - 2N + it$  with  $|t| \leq T_N$ , and use (5.3). Then

$$|(2\pi x)^{-s}| = (2\pi x)^{2N - \frac{1}{2}},$$

and (5.7), (5.8) bound the two sine factors. Since

$$\operatorname{Re}(1 - s) = 2N + \frac{1}{2} \geq \frac{5}{2},$$

Lemma 5.1 gives

$$(5.18) \quad |\Gamma(1-s)| \geq \Gamma(2N + \tfrac{1}{2}) (1+t^2)^{-1/2} e^{-\pi|t|/2},$$

the factor  $1+t^2$  arising from Lemma 5.1 applied with real part

$$1-\sigma = 2N + \tfrac{1}{2} \geq 1,$$

in which

$$1 + \frac{t^2}{(1-\sigma)^2} \leq 1+t^2,$$

and (1.11) applied at  $\operatorname{Re}(1-s) \geq \frac{5}{2}$  gives

$$|\zeta(1-s)| \geq \frac{\zeta(5)}{\zeta(5/2)} =: c_0 > 0.$$

Assembling these bounds in (5.3),

$$(5.19) \quad |F(s)| \leq \frac{\pi^2 (2\pi x)^{2N-\frac{1}{2}} \cdot 2e^{-\pi|t|} \cdot \sqrt{2} \cdot (1+t^2) e^{\pi|t|}}{\Gamma(2N + \tfrac{1}{2})^2 c_0} \\ = C_2 \frac{(2\pi x)^{2N} (1+t^2)}{\Gamma(2N + \tfrac{1}{2})^2}.$$

The exponential factors cancel exactly; this cancellation, which no fixed- $\sigma$  asymptotic can deliver, is the point of Lemma 5.1. Integrating over  $|t| \leq T_N \leq N+1$ ,

$$(5.20) \quad \left| \int_{\text{left edge}} F(s) ds \right| \leq C_3 \frac{(2\pi x)^{2N} N^3}{\Gamma(2N + \tfrac{1}{2})^2} \longrightarrow 0 \quad (N \rightarrow \infty),$$

superexponentially, for every fixed  $x > 0$ , since the factorial in the denominator dominates any fixed exponential.

*Segments  $\mathcal{C}_2$  and  $\mathcal{C}_4$ : the horizontal edges (red and orange in Figure 1).* We work on the lines  $t = \pm T_N$ , with the upper sign on  $\mathcal{C}_2$  and the lower sign on  $\mathcal{C}_4$ . Split each segment at  $\sigma = -1$  into a near part,  $-1 \leq \sigma \leq \sigma_0$ , and a far part,  $\frac{1}{2} - 2N \leq \sigma \leq -1$ .

For  $-1 \leq \sigma \leq \sigma_0$ , Lemma 5.2 gives

$$\frac{1}{|\zeta(s)|} \leq T_N^{A_0};$$

the Stirling estimate ([7, Section 5.11]), which is uniform for  $\sigma$  in a compact interval,

$$(5.21) \quad |\Gamma(\sigma \pm iT)| \leq C_4 T^{\sigma_0 - \frac{1}{2}} e^{-\pi T/2} \quad (-1 \leq \sigma \leq \sigma_0, T \geq 1),$$



together with  $x^{-\sigma} \leq \max(x^{-3/2}, x)$ , yields

$$(5.22) \quad \left| \int_{-1}^{\sigma_0} F(\sigma \pm iT_N) d\sigma \right| \leq C_5(x) T_N^{A_0 + \sigma_0 - \frac{1}{2}} e^{-\pi T_N/2} \longrightarrow 0 \quad (N \rightarrow \infty),$$

the exponential factor  $e^{-\pi T_N/2}$  crushing the fixed power of  $T_N$ .

For  $\frac{1}{2} - 2N \leq \sigma \leq -1$  we use (5.3) again. By (5.9) and (5.10), by Lemma 5.1 applied at  $\operatorname{Re}(1-s) = 1-\sigma \geq 2$ ,

$$(5.23) \quad |\Gamma(1-s)| \geq \Gamma(1-\sigma) (1+T_N^2)^{-1/2} e^{-\pi T_N/2},$$

the factor  $1+T_N^2$  again coming from Lemma 5.1 with real part  $1-\sigma \geq 2$ , in which

$$1 + \frac{t^2}{(1-\sigma)^2} \leq 1 + t^2 \leq 1 + T_N^2 \quad (|t| \leq T_N);$$

and by  $|\zeta(1-s)| \geq \zeta(4)/\zeta(2)$  from (1.11), we obtain

$$(5.24) \quad |F(\sigma \pm iT_N)| \leq C_6 \frac{(2\pi x)^{-\sigma} (1+T_N^2)}{\Gamma(1-\sigma)^2} e^{-\pi T_N - \frac{\pi T_N}{2} + \pi T_N} \\ = C_6 \frac{(2\pi x)^{-\sigma} (1+T_N^2)}{\Gamma(1-\sigma)^2} e^{-\pi T_N/2},$$

with the exponentials again cancelling except for the decisive factor  $e^{-\pi T_N/2}$ . To integrate the bound (5.24) over  $\sigma$ , we enlarge the range  $\frac{1}{2} - 2N \leq \sigma \leq -1$  of the horizontal segment to the full half-line  $\sigma \leq -1$ , which only increases the integral since the integrand is positive; substituting  $u = 1-\sigma$ , so that  $u$  runs over  $[2, \infty)$ , produces the quantity

$$(5.25) \quad W(x) := \int_{-\infty}^{-1} \frac{(2\pi x)^{-\sigma}}{\Gamma(1-\sigma)^2} d\sigma = \int_2^{\infty} \frac{(2\pi x)^{u-1}}{\Gamma(u)^2} du < \infty,$$

finite for every  $x > 0$  because, by (H.1) or Stirling's formula,  $\Gamma(u)^2$  grows superexponentially while  $(2\pi x)^{u-1}$  grows only exponentially in  $u$ . Hence the far part of each horizontal edge is

$$(5.26) \quad \left| \int_{\frac{1}{2}-2N}^{-1} F(\sigma \pm iT_N) d\sigma \right| \leq C_6 W(x) (1+T_N^2) e^{-\pi T_N/2} \longrightarrow 0 \quad (N \rightarrow \infty).$$

*Conclusion.* Letting  $N \rightarrow \infty$  in (5.16): the contour integral on the left tends to  $\Phi(e^{-x})$ : the segment  $\mathcal{C}_1$  contributes the integral by (5.17), the segment  $\mathcal{C}_3$  tends to zero by (5.20), and the segments  $\mathcal{C}_2$  and  $\mathcal{C}_4$  tend to zero by (5.22) and (5.26); the two series over the trivial poles converge absolutely for every fixed  $x > 0$ , their terms being crushed by the squared factorials in (3.8) and (4.13) (Lemma 6.1 below), and their

sums are  $\pi\kappa(x)$  by (3.12) and  $2\lambda(x) + 4v(x) - 2\beta(x) \log(2\pi x)$  by (4.14). Consequently the remaining term of (5.16), the symmetric partial sum over the nontrivial zeros with  $|\gamma| < T_N$ , converges as  $N \rightarrow \infty$  as well, and its limit is determined by the other four terms; rearranging gives exactly (1.18). This proves Theorem 1.1.  $\square$

*Remark 5.5* (Multiple zeros). Hypothesis (S) enters only through the closed form (2.1). Without it, each zero  $\rho$  of multiplicity  $m_\rho$  contributes

$$(5.27) \quad \frac{1}{(m_\rho - 1)!} \lim_{s \rightarrow \rho} \frac{d^{m_\rho-1}}{ds^{m_\rho-1}} \left( (s - \rho)^{m_\rho} \frac{\Gamma(s) x^{-s}}{\zeta(s)} \right),$$

and Theorem 1.1 holds unconditionally in this form, exactly as in the unconditional treatment of the Mertens formula by Bartz [3].

## 6. THE SPECIAL FUNCTIONS $\kappa, \beta, \lambda, v$

**Lemma 6.1.** *The four power series defining  $\kappa(x)$ ,  $\beta(x)$ ,  $\lambda(x)$  and  $v(x)$  have infinite radius of convergence; the four functions are entire.*

*Proof.* Write each of the four series as  $\sum_n a_n x^n$ , with  $n = 2m + 1$  for  $\kappa$  and  $n = 2m$  for  $\beta, \lambda, v$ , the remaining coefficients being zero. By (1.16) and (1.17), every nonzero coefficient obeys

$$(6.1) \quad |a_n| \leq C \frac{(n+1)(2\pi)^n}{(n!)^2}$$

with an absolute constant  $C$ . Indeed, the zeta factors lie in a fixed bounded set, since  $\zeta(2m+1) > 1$ ,  $\zeta(2m+2) > 1$ , and

$$|\zeta'(2m+1)| \leq \sum_{k=2}^{\infty} \frac{\log k}{k^3} < \infty;$$

and the derivative factor in  $v$  is controlled by the inequality

$$(6.2) \quad \begin{aligned} \Gamma'(s) &= \frac{d}{ds} \int_0^\infty t^{s-1} e^{-t} dt = \int_0^\infty t^{s-1} (\log t) e^{-t} dt \\ &< \int_0^\infty t^s e^{-t} dt = \Gamma(s+1), \end{aligned}$$

valid for every real  $s > 0$ : the differentiation under the integral sign is the standard theorem for parameter integrals, the  $s$ -derivative of the integrand being locally dominated by an integrable function; and the strict inequality is the pointwise bound  $\log t < t$  for  $t > 0$ , integrated against the positive weight  $t^{s-1} e^{-t}$ . Applied with  $s = 2m + 1$  this gives

$$\frac{\Gamma(2m+1)}{[(2m)!]^3} < \frac{\Gamma(2m+2)}{[(2m)!]^3} = \frac{2m+1}{[(2m)!]^2},$$

which is (6.1) for  $v$ . Now apply the Cauchy–Hadamard formula: the radius of convergence is  $R = 1/\limsup_{n \rightarrow \infty} |a_n|^{1/n}$ , the  $\limsup$  being unaffected by the vanishing coefficients. By the elementary inequality

$$n! \geq \left(\frac{n}{e}\right)^n, \quad \text{which follows from} \quad e^n = \sum_{j=0}^{\infty} \frac{n^j}{j!} \geq \frac{n^n}{n!},$$

the bound (6.1) gives

$$\begin{aligned} |a_n|^{1/n} &\leq \frac{2\pi (C(n+1))^{1/n}}{(n!)^{2/n}} \leq \frac{2\pi e^2 (C(n+1))^{1/n}}{n^2} \\ &\rightarrow 0 \quad (n \rightarrow \infty), \end{aligned}$$

so  $\limsup_{n \rightarrow \infty} |a_n|^{1/n} = 0$  and the radius of convergence is infinite for all four series.  $\square$

We now prove Theorem 1.2. The Bessel function of the first kind of order zero is

$$(6.3) \quad J_0(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\frac{z}{2}\right)^{2n}$$

(see [13, Chapter 5] or [7, Section 10.2]). Substituting the rotated argument  $z = 2e^{-i\pi/4}\sqrt{y}$ , for which  $(z/2)^2 = e^{-i\pi/2}y = -iy$ , we obtain the identity

$$(6.4) \quad J_0(2e^{-i\pi/4}\sqrt{y}) = \sum_{n=0}^{\infty} \frac{(-1)^n(-i)^n}{(n!)^2} y^n = \sum_{n=0}^{\infty} \frac{i^n}{(n!)^2} y^n \quad (y \geq 0),$$

whose real and imaginary parts are precisely the two lacunary series that arise from the trivial zeros:

$$(6.5) \quad \operatorname{Re} J_0(2e^{-i\pi/4}\sqrt{y}) = \sum_{m=0}^{\infty} \frac{(-1)^m y^{2m}}{[(2m)!]^2},$$

$$(6.6) \quad \operatorname{Im} J_0(2e^{-i\pi/4}\sqrt{y}) = \sum_{m=0}^{\infty} \frac{(-1)^m y^{2m+1}}{[(2m+1)!]^2}.$$

(By the classical relation  $\operatorname{ber}(z) + i \operatorname{bei}(z) = J_0(z e^{3\pi i/4})$  and the evenness of  $J_0$ , the real and imaginary parts in (6.5) and (6.6) are precisely the Kelvin functions  $\operatorname{ber}(2\sqrt{y})$  and  $\operatorname{bei}(2\sqrt{y})$ ; see [7, Section 10.61].)

*Proof of Theorem 1.2.* Expanding  $1/\zeta(2m+2) = \sum_{k \geq 1} \mu(k)k^{-2m-2}$  in the definition (1.16) of  $\kappa$  and interchanging the order of summation,

$$(6.7) \quad \begin{aligned} \kappa(x) &= \sum_{k=1}^{\infty} \frac{\mu(k)}{k} \sum_{m=0}^{\infty} \frac{(-1)^m}{[(2m+1)!]^2} \left(\frac{2\pi x}{k}\right)^{2m+1} \\ &= \sum_{k=1}^{\infty} \frac{\mu(k)}{k} \operatorname{Im} J_0\left(2e^{-i\pi/4} \sqrt{\frac{2\pi x}{k}}\right), \end{aligned}$$

by (6.6) with  $y = 2\pi x/k$ . The interchange is justified by Tonelli's theorem: putting  $y_k = 2\pi x/k$  and  $g(y) = \sum_{m \geq 0} y^{2m+1}/[(2m+1)!]^2$ , one has  $g(y) \leq y e^y$ , so

$$(6.8) \quad \sum_{k=1}^{\infty} \frac{1}{k} g(y_k) \leq \sum_{k \leq 2\pi x} \frac{g(y_k)}{k} + \sum_{k > 2\pi x} \frac{y_k e^{y_k}}{k} \leq C(x) + 2\pi e x \sum_{k > 2\pi x} \frac{1}{k^2} < \infty.$$

Identically,  $1/\zeta(2m+1) = \sum_k \mu(k)k^{-2m-1}$  gives

$$(6.9) \quad \begin{aligned} \beta(x) &= \sum_{k=1}^{\infty} \frac{\mu(k)}{k} \sum_{m=1}^{\infty} \frac{(-1)^m}{[(2m)!]^2} \left(\frac{2\pi x}{k}\right)^{2m} \\ &= \sum_{k=1}^{\infty} \frac{\mu(k)}{k} \left( \operatorname{Re} J_0\left(2e^{-i\pi/4} \sqrt{\frac{2\pi x}{k}}\right) - 1 \right), \end{aligned}$$

where the inner sum starting at  $m = 1$  is exactly  $\operatorname{Re} J_0 - 1$ ; the corresponding double sum has terms  $O(x^2/k^3)$ , since  $\operatorname{Re} J_0(2e^{-i\pi/4} \sqrt{y}) - 1 = -\frac{y^2}{4} + O(y^4)$  as  $y \rightarrow 0$ , so the interchange is again absolute. Adding  $i$  times (6.7) to (6.9) yields the combined identity (1.19). It is worth noting that the subtraction of 1 inside (6.9) is exactly what removes the conditionally convergent series  $\sum_k \mu(k)/k$  from the picture, so no appeal to the prime number theorem is needed anywhere.

For  $\lambda$  we use the Dirichlet series of the derivative of  $1/\zeta$ : for  $\operatorname{Re} s > 1$ ,

$$(6.10) \quad \frac{\zeta'(s)}{\zeta(s)^2} = -\left(\frac{1}{\zeta}\right)'(s) = \sum_{k=1}^{\infty} \frac{\mu(k) \log k}{k^s},$$

obtained by differentiating (1.2) term by term. Substituting (6.10) at  $s = 2m+1$  into the definition (1.17) of  $\lambda$  and interchanging as before, with terms now  $O(x^2 \log k/k^3)$ ,

$$(6.11) \quad \lambda(x) = \sum_{k=1}^{\infty} \frac{\mu(k) \log k}{k} \left( \operatorname{Re} J_0\left(2e^{-i\pi/4} \sqrt{\frac{2\pi x}{k}}\right) - 1 \right),$$

which is (1.20). (Here the  $k = 1$  term vanishes because  $\log 1 = 0$ , and no constant needs to be subtracted.) Together with Lemma 6.1 this proves Theorem 1.2.  $\square$

## 7. THE SUM OVER THE ZEROS

**Theorem 7.1.** *Assume Hypothesis (S). For every  $0 < x < \infty$  the symmetric limit*

$$(7.1) \quad \sum_{\rho} \frac{\Gamma(\rho) x^{-\rho}}{\zeta'(\rho)} = \lim_{\nu \rightarrow \infty} \sum_{|\gamma| < T_{\nu}} \frac{\Gamma(\rho) x^{-\rho}}{\zeta'(\rho)}$$

*exists and is a real number.*

*Proof.* Existence of the limit is a byproduct of the proof of Theorem 1.1: in (5.16) every other term converges as  $N \rightarrow \infty$ . Reality follows from the conjugate symmetry of the zeros:  $\rho$  is a zero if and only if  $\bar{\rho}$  is a zero, both are simple together, and since  $\overline{\Gamma(\bar{s})} = \Gamma(s)$ ,  $\overline{\zeta'(\bar{s})} = \zeta'(s)$  and  $x > 0$ ,

$$(7.2) \quad \frac{\Gamma(\bar{\rho}) x^{-\bar{\rho}}}{\zeta'(\bar{\rho})} = \overline{\left( \frac{\Gamma(\rho) x^{-\rho}}{\zeta'(\rho)} \right)},$$

so each symmetric partial sum, which contains the zeros in conjugate pairs, equals twice the real part of the sum over the zeros with  $\gamma > 0$ .  $\square$

For an absolutely convergent version we introduce the following hypothesis.

**Hypothesis (H).**  $\sum_{\rho} \frac{|\Gamma(\rho)|}{|\zeta'(\rho)|} < \infty.$

Since, by (1.7) with the constant uniform for  $\beta_{\rho} \in [0, 1]$ ,

$$|\Gamma(\rho)| \ll e^{-\pi|\gamma|/2} (1 + |\gamma|)^{\beta_{\rho} - \frac{1}{2}},$$

and the number of zeros with  $|\gamma| \leq T$  is  $O(T \log T)$  ([16, Chapter IX]), Hypothesis (H) holds as soon as, for a single fixed  $\theta < \pi/2$ ,

$$\frac{1}{|\zeta'(\rho)|} \ll e^{\theta|\gamma|} \quad \text{over the nontrivial zeros.}$$

This is an extraordinarily weak requirement: the conjectures on negative moments of  $\zeta'$  at the zeros, going back to Gonek [9] and Hejhal [11], predict polynomial bounds. We emphasize, however, that Hypothesis (S) alone does not imply (H), since simplicity does not by itself exclude zeros at which  $|\zeta'(\rho)|$  is extremely small.

The sum (7.1) can also be written as a sum over the zeros in the upper half of the critical strip only. This follows from the fact that if  $\rho$  is a zero of zeta then  $1 - \rho$  is also a zero of zeta, that the map  $\rho \mapsto 1 - \rho$  exchanges the upper and lower halves of the critical strip, and that the first zero of zeta has imaginary part bigger than 1 (indeed  $\gamma_1 = 14.134\dots$ ; see [16, Chapter XV]):

$$(7.3) \quad \sum_{\rho} \frac{\Gamma(\rho) x^{-\rho}}{\zeta'(\rho)} = \sum_{\operatorname{Im} \rho > 0} \left( \frac{\Gamma(\rho) x^{-\rho}}{\zeta'(\rho)} + \frac{\Gamma(1 - \rho) x^{-(1-\rho)}}{\zeta'(1 - \rho)} \right).$$

Using the functional equation (3.4) we see, differentiating and evaluating at  $s = \rho$  where  $\zeta(1 - \rho) = 0$ ,

$$(7.4) \quad \zeta'(\rho) = \left( 2^s \pi^{s-1} \sin(\tfrac{1}{2}\pi s) \Gamma(1 - s) \right)' \Big|_{s=\rho} \underbrace{\zeta(1 - \rho)}_{=0} - 2^\rho \pi^{\rho-1} \sin(\tfrac{1}{2}\pi \rho) \Gamma(1 - \rho) \zeta'(1 - \rho),$$

therefore

$$(7.5) \quad \zeta'(\rho) = -2^\rho \pi^{\rho-1} \sin(\tfrac{1}{2}\pi \rho) \Gamma(1 - \rho) \zeta'(1 - \rho),$$

and hence

$$(7.6) \quad \zeta'(1 - \rho) = -\frac{\zeta'(\rho) 2^{-\rho} \pi^{1-\rho}}{\Gamma(1 - \rho) \sin(\tfrac{1}{2}\pi \rho)}.$$

Substituting (7.6), and keeping careful track of the minus sign,

$$(7.7) \quad \begin{aligned} \frac{\Gamma(1 - \rho) x^{-(1-\rho)}}{\zeta'(1 - \rho)} &= -\frac{\Gamma(1 - \rho)^2 \sin(\tfrac{1}{2}\pi \rho) 2^\rho \pi^{\rho-1}}{\zeta'(\rho)} x^{\rho-1} \\ &= -\pi \frac{(2\pi)^\rho \sin(\tfrac{1}{2}\pi \rho)}{\zeta'(\rho) \Gamma(\rho)^2 \sin^2(\pi \rho)} x^{\rho-1}, \end{aligned}$$

where in the last step we used  $\Gamma(1 - \rho)^2 = \pi^2 / (\Gamma(\rho)^2 \sin^2(\pi \rho))$  from (5.1) and  $2\pi^2 2^{\rho-1} \pi^{\rho-1} = \pi(2\pi)^\rho$ . Therefore

$$(7.8) \quad \begin{aligned} \sum_{\rho} \frac{\Gamma(\rho) x^{-\rho}}{\zeta'(\rho)} &= \sum_{\operatorname{Im} \rho > 0} \left[ \frac{\Gamma(\rho)}{\zeta'(\rho)} x^{-\rho} - \frac{2(2\pi)^{\rho-1} \Gamma(1 - \rho)^2 \sin(\tfrac{1}{2}\pi \rho)}{\zeta'(\rho)} x^{\rho-1} \right], \end{aligned}$$

and equivalently

$$(7.9) \quad \sum_{\rho} \frac{\Gamma(\rho) x^{-\rho}}{\zeta'(\rho)} = \sum_{\operatorname{Im} \rho > 0} \left[ \frac{\Gamma(\rho)}{\zeta'(\rho)} x^{-\rho} - \pi \frac{(2\pi)^\rho \sin(\tfrac{1}{2}\pi \rho)}{\zeta'(\rho) \Gamma(\rho)^2 \sin^2(\pi \rho)} x^{\rho-1} \right].$$

## 8. A CRITERION FOR THE RIEMANN HYPOTHESIS

Here, by  $f(x) = O(g(x))$  as  $x \rightarrow 0^+$  we mean that there exists  $\delta > 0$  such that  $|f(x)/g(x)|$  is bounded for  $0 < x < \delta$ .

*Proof of Theorem 1.3(b).* Assume RH, Hypothesis (S) and Hypothesis (H). By Theorem 1.1, and since on RH every zero is  $\rho = \frac{1}{2} + i\gamma$  so that  $x^{-\rho} = x^{-1/2}x^{-i\gamma}$  with  $|x^{-i\gamma}| = 1$ ,

$$(8.1) \quad \sum_{n=1}^{\infty} \mu(n)e^{-nx} = x^{-\frac{1}{2}} \sum_{\rho} \frac{\Gamma(\rho)}{\zeta'(\rho)} x^{-i\gamma} + \pi\kappa(x) + 2\lambda(x) + 4v(x) - 2\beta(x) \log(2\pi x) - 2.$$

Since  $\sum_{\rho} |\Gamma(\rho)/\zeta'(\rho)|$  is convergent by Hypothesis (H), we have

$$(8.2) \quad \left| \sum_{\rho} \frac{\Gamma(\rho)}{\zeta'(\rho)} x^{-i\gamma} \right| \leq \sum_{\rho} \left| \frac{\Gamma(\rho)}{\zeta'(\rho)} \right| < \infty,$$

uniformly in  $x$ . As  $x \rightarrow 0^+$ , the definitions (1.16) and (1.17) give

$$\kappa(x) = O(x), \quad \beta(x), \lambda(x), v(x) = O(x^2),$$

each series beginning at the power  $x$  or  $x^2$  respectively and converging on compact sets, and hence

$$\beta(x) \log(2\pi x) = O(x^2 \log(x));$$

all trivial contributions are therefore  $O(1)$ . Combining,

$$\sum_n \mu(n)e^{-nx} = O(x^{-1/2}).$$

□

*Proof of Theorem 1.3(a).* Assume (1.21), say  $|\Phi(e^{-x})| \leq Cx^{-1/2}$  for  $0 < x \leq 1$ . For  $x \geq 1$  we have the trivial bound

$$|\Phi(e^{-x})| \leq \sum_{n \geq 1} e^{-nx} = (e^x - 1)^{-1} \leq 2e^{-x}.$$

Consequently the Mellin integral

$$(8.3) \quad G(s) := \int_0^{\infty} \Phi(e^{-x}) x^{s-1} dx$$

converges absolutely and locally uniformly in the half-plane  $\operatorname{Re} s > \frac{1}{2}$ ; indeed

$$\int_0^1 x^{\sigma-3/2} dx < \infty$$

precisely for  $\sigma > \frac{1}{2}$ , and the tail is dominated by

$$2 \int_1^\infty e^{-x} x^{\sigma-1} dx.$$

Hence  $G$  is analytic in  $\operatorname{Re} s > \frac{1}{2}$ . On the other hand, for  $\operatorname{Re} s > 1$  the computation (1.5), justified there by Tonelli's theorem, gives  $G(s) = \Gamma(s)/\zeta(s)$ . By the uniqueness of analytic continuation,  $\Gamma(s)/\zeta(s)$  extends to an analytic function in the whole half-plane  $\operatorname{Re} s > \frac{1}{2}$ . Now suppose  $\zeta$  had a zero  $\rho_0$  with  $\operatorname{Re} \rho_0 > \frac{1}{2}$ . Since  $\Gamma$  is zero-free in the complex plane,  $\Gamma(s)/\zeta(s)$  would have a pole at  $\rho_0$ , contradicting analyticity. (At  $s = 1$  there is no obstruction: the pole of  $\zeta$  makes  $\Gamma/\zeta$  vanish there.) Therefore  $\zeta$  has no zeros with real part exceeding  $\frac{1}{2}$ , and by the functional equation (3.4) it has none with real part less than  $\frac{1}{2}$  in the critical strip either; all nontrivial zeros lie on the critical line. Note that neither Hypothesis (S) nor Hypothesis (H) was used.  $\square$

## 9. CONCLUDING REMARKS

- (1) The structural novelty of (1.18) relative to the Mertens formula (1.22) is the collision of the poles of  $\Gamma(s)$  with the trivial zeros, which produces the double poles of Section 4 and the term  $-2\beta(x) \log(2\pi x)$ . Any kernel whose Mellin transform has poles at the negative even integers will exhibit the same phenomenon.
- (2) By Remark 5.5, the explicit formula holds unconditionally with the multiplicity-adjusted zero terms (5.27); Hypothesis (S) yields the closed form (2.1).
- (3) The Laplace kernel is the member  $\alpha = 1$  of the family  $e^{-n^\alpha x}$ , whose Mellin transforms are again Gamma functions up to scaling; the corresponding explicit formulas of general order, in which the logarithmic terms turn out to be present exactly for rational  $\alpha$  and absent for irrational  $\alpha$ , are developed in a separate continuation of this paper.
- (4) The arguments adapt to Dirichlet  $L$ -functions, with  $\mu(n)\chi(n)$  in place of  $\mu(n)$  and the trivial zeros shifted according to the parity of the character; in view of [8] this suggests character-twisted Möbius–Bessel expansions.
- (5) It would be interesting to determine whether Hypothesis (H) can be removed from Theorem 1.3(b) by smoothing the zero sum, and to relate  $\kappa$  analytically to the Riesz function [15] and to the Hardy–Littlewood series [10], whose kernels generate Bessel functions of unrotated argument.



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## APPENDIX A. SINE ESTIMATES ON THE CONTOURS

This and the following appendices record, at the level of detail of a formal verification, the auxiliary computations supporting the main text. Throughout,  $m, N \in \{1, 2, 3, \dots\}$ ,  $x > 0$ ,  $\gamma_e$  is Euler's constant, and  $\psi = \Gamma'/\Gamma$ .

We derive the four estimates (5.7), (5.8), (5.9), (5.10). On the vertical line  $s = \frac{1}{2} - 2N + it$ ,

$$(A.1) \quad \pi\left(\frac{1}{2} - 2N + it\right) = \frac{\pi}{2} - 2\pi N + i\pi t,$$

and by  $\sin(\frac{\pi}{2} + z) = \cos z$ , the  $2\pi$ -periodicity of sine, and  $\cos(i\pi t) = \cosh(\pi t)$ ,

$$(A.2) \quad \sin\left(\pi\left(\frac{1}{2} - 2N + it\right)\right) = \cos(i\pi t) = \cosh(\pi t),$$

so that, by  $\cosh u = \frac{1}{2}(e^u + e^{-u}) \geq \frac{1}{2}e^{|u|}$ ,

$$(A.3) \quad \left| \sin\left(\pi\left(\frac{1}{2} - 2N + it\right)\right) \right| = \cosh(\pi t) \geq \frac{1}{2}e^{\pi|t|},$$

which is (5.7). For the half-angle,  $\frac{\pi}{2}\left(\frac{1}{2} - 2N + it\right) = \frac{\pi}{4} - \pi N + i\frac{\pi t}{2}$ , and the identity  $|\sin(a + ib)|^2 = \sin^2 a + \sinh^2 b$  (which is (5.6)) gives, with

$$\sin^2\left(\frac{\pi}{4} - \pi N\right) = \frac{1}{2} \text{ and } 1 + 2\sinh^2 u = \cosh(2u),$$

$$(A.4) \quad \left| \sin\left(\frac{\pi}{2}\left(\frac{1}{2} - 2N + it\right)\right) \right|^2 = \frac{1}{2} + \sinh^2\left(\frac{\pi t}{2}\right) = \frac{1}{2} \cosh(\pi t) \geq \frac{1}{2},$$

which is (5.8). On the horizontal lines  $s = \sigma \pm iT$  with  $T \geq 1$ , the same identity with  $a = \pi\sigma$ ,  $b = \pm\pi T$  gives

$$(A.5) \quad |\sin(\pi(\sigma \pm iT))|^2 = \sin^2(\pi\sigma) + \sinh^2(\pi T) \geq \sinh^2(\pi T),$$

and, since  $T \geq 1$ ,

$$\sinh(\pi T) = \frac{1}{2}(e^{\pi T} - e^{-\pi T}) \geq \frac{1}{2}e^{\pi T}(1 - e^{-2\pi}) \geq \frac{1}{4}e^{\pi T},$$

which is (5.9). Identically,

$$\left| \sin\left(\frac{\pi}{2}(\sigma \pm iT)\right) \right| \geq \sinh\left(\frac{\pi T}{2}\right) \geq \frac{1}{2}e^{\pi T/2}(1 - e^{-\pi}) \geq \frac{1}{4}e^{\pi T/2},$$

which is (5.10).

## APPENDIX B. ZETA VALUES USED IN THE MAIN TEXT

We record  $\zeta(0) = -\frac{1}{2}$ , the trivial zeros  $\zeta(-2m) = 0$ , and Euler's evaluation (3.2), equivalently

$$\zeta(2m) = \frac{|B_{2m}| (2\pi)^{2m}}{2(2m)!};$$

for example,

$$\zeta(2) = \frac{\pi^2}{6}, \quad \zeta(4) = \frac{\pi^4}{90}, \quad \zeta(6) = \frac{\pi^6}{945}.$$

All are classical ([16, Chapter II]);  $\zeta(0) = -\frac{1}{2}$  also follows from the functional equation (3.4) by letting  $s \rightarrow 0$  and using  $\zeta(1-s) \sim -1/s$  together with  $\sin(\frac{\pi s}{2}) \sim \frac{\pi s}{2}$ :

$$(B.1) \quad \zeta(0) = \lim_{s \rightarrow 0} 2^s \pi^{s-1} \cdot \frac{\pi s}{2} \cdot \Gamma(1-s) \cdot \left(-\frac{1}{s}\right) = -\frac{1}{2}.$$

## APPENDIX C. LAURENT EXPANSION OF THE GAMMA FUNCTION AT $s = -n$

We prove (2.5): for every integer  $n \geq 0$ , as  $s \rightarrow -n$ , with  $w = s + n$ ,

$$(C.1) \quad \Gamma(s) = \frac{(-1)^n}{n!} \left( \frac{1}{w} + \psi(n+1) + O(w) \right).$$

Iterating the recurrence  $\Gamma(z+1) = z\Gamma(z)$  exactly  $n+1$  times,

$$(C.2) \quad \Gamma(s) = \frac{\Gamma(s+n+1)}{s(s+1)\cdots(s+n)} = \frac{\Gamma(1+w)}{(w-n)(w-n+1)\cdots(w-1)w}.$$

The numerator expands as  $\Gamma(1+w) = 1 - \gamma_e w + O(w^2)$ , by  $\Gamma'(1) = -\gamma_e$ . The product of the  $n$  nonvanishing factors in the denominator is

$$(C.3) \quad \prod_{j=1}^n (w-j) = (-1)^n n! \prod_{j=1}^n \left(1 - \frac{w}{j}\right) = (-1)^n n! (1 - H_n w + O(w^2)),$$

where

$$H_n = \sum_{j=1}^n \frac{1}{j}$$

is the  $n$ -th harmonic number, with the empty-sum convention  $H_0 = 0$ . Therefore

$$(C.4) \quad \begin{aligned} \Gamma(s) &= \frac{1 - \gamma_e w + O(w^2)}{(-1)^n n! w (1 - H_n w + O(w^2))} \\ &= \frac{(-1)^n}{n!} \cdot \frac{1}{w} (1 + (H_n - \gamma_e)w + O(w^2)), \end{aligned}$$

and since  $\psi(n+1) = H_n - \gamma_e$  ([7, Section 5.4]), this is (C.1). In the application to the trivial zeros we take  $n = 2m$ , where  $(-1)^{2m} = 1$ :

$$(C.5) \quad \Gamma(s) = \frac{1}{(2m)!} \left( \frac{1}{w} + \psi(2m+1) + O(w) \right) \quad (w = s + 2m \rightarrow 0).$$

#### APPENDIX D. EXPANSION OF $1/\zeta$ AND OF $x^{-s}$ AT A TRIVIAL ZERO

Since  $\zeta(-2m) = 0$  and the trivial zeros are simple,  $\zeta'(-2m) \neq 0$ , and Taylor expansion at  $-2m$  gives, with  $w = s + 2m$ ,

$$(D.1) \quad \zeta(s) = \zeta'(-2m) w \left( 1 + \frac{\zeta''(-2m)}{2\zeta'(-2m)} w + O(w^2) \right),$$

whence, inverting the unit factor by the geometric series,

$$(D.2) \quad \frac{1}{\zeta(s)} = \frac{1}{\zeta'(-2m) w} \left( 1 - \frac{\zeta''(-2m)}{2\zeta'(-2m)} w + O(w^2) \right).$$

Also  $x^{-s} = x^{2m-w} = x^{2m} e^{-w \log x}$ , so

$$(D.3) \quad x^{-s} = x^{2m} (1 - w \log x + O(w^2)).$$

APPENDIX E. DIRECT LAURENT COMPUTATION OF THE RESIDUE  
AT  $s = -2m$

Multiplying (C.5), (D.2) and (D.3),

$$(E.1) \quad \frac{\Gamma(s) x^{-s}}{\zeta(s)} = \frac{x^{2m}}{(2m)! \zeta'(-2m)} \cdot \frac{1}{w^2} \left( 1 + \psi(2m+1) w + O(w^2) \right) \\ \times \left( 1 - \frac{\zeta''(-2m)}{2 \zeta'(-2m)} w + O(w^2) \right) \left( 1 - w \log x + O(w^2) \right).$$

The coefficient of  $w$  in the product of the three unit factors is  $\psi(2m+1) - \frac{\zeta''(-2m)}{2\zeta'(-2m)} - \log x$ , so the coefficient of  $w^{-1}$  in (E.1), that is the residue, equals

$$(E.2) \quad \operatorname{Res}_{s=-2m} \frac{\Gamma(s) x^{-s}}{\zeta(s)} = \frac{x^{2m}}{(2m)! \zeta'(-2m)} \left( \psi(2m+1) - \frac{\zeta''(-2m)}{2 \zeta'(-2m)} - \log x \right),$$

which is the residue whose agreement with (4.13) is verified in Appendix G.

APPENDIX F. FUNCTIONAL-EQUATION EVALUATION OF  $\zeta'(-2m)$   
AND  $\zeta''(-2m)$

Write the functional equation (3.4) as the factorization

$$(F.1) \quad \zeta(s) = h(s) \sin\left(\frac{\pi s}{2}\right), \quad h(s) = 2^s \pi^{s-1} \Gamma(1-s) \zeta(1-s),$$

where  $h$  is analytic and nonzero in a neighbourhood of  $s = -2m$ : indeed  $\Gamma(1-s)$  and  $\zeta(1-s)$  are analytic and nonzero at  $s = -2m$ , their arguments being  $2m+1 > 1$ . With  $w = s + 2m$ ,

$$(F.2) \quad \sin\left(\frac{\pi s}{2}\right) = \sin\left(\frac{\pi w}{2} - \pi m\right) = (-1)^m \sin\left(\frac{\pi w}{2}\right) = (-1)^m \left(\frac{\pi w}{2} + O(w^3)\right),$$

and  $h(s) = h(-2m) + h'(-2m) w + O(w^2)$ , so

$$(F.3) \quad \zeta(s) = (-1)^m \frac{\pi}{2} h(-2m) w + (-1)^m \frac{\pi}{2} h'(-2m) w^2 + O(w^3).$$

Comparing with (D.1) coefficient by coefficient,

$$(F.4) \quad \zeta'(-2m) = (-1)^m \frac{\pi}{2} h(-2m), \quad \frac{\zeta''(-2m)}{2} = (-1)^m \frac{\pi}{2} h'(-2m), \\ \frac{\zeta''(-2m)}{2 \zeta'(-2m)} = \frac{h'(-2m)}{h(-2m)}.$$

Since

$$h(-2m) = 2^{-2m} \pi^{-2m-1} (2m)! \zeta(2m+1),$$

the first identity gives the closed form

$$(F.5) \quad \zeta'(-2m) = \frac{(-1)^m (2m)! \zeta(2m+1)}{2 (2\pi)^{2m}}.$$

For the logarithmic derivative,  $\log h(s) = s \log 2 + (s-1) \log \pi + \log \Gamma(1-s) + \log \zeta(1-s)$  on a simply connected neighbourhood of  $-2m$  avoiding zeros of  $h$ , so

$$(F.6) \quad \frac{h'(s)}{h(s)} = \log(2\pi) - \psi(1-s) - \frac{\zeta'(1-s)}{\zeta(1-s)},$$

and evaluating at  $s = -2m$  and inserting into (F.4) gives

$$(F.7) \quad \frac{\zeta''(-2m)}{2\zeta'(-2m)} = \log(2\pi) - \psi(2m+1) - \frac{\zeta'(2m+1)}{\zeta(2m+1)}.$$

#### APPENDIX G. AGREEMENT OF THE TWO RESIDUE COMPUTATIONS

Substituting (F.7) into (E.2), the bracket becomes

$$(G.1) \quad \begin{aligned} \psi(2m+1) - \left( \log(2\pi) - \psi(2m+1) - \frac{\zeta'(2m+1)}{\zeta(2m+1)} \right) - \log x \\ = 2\psi(2m+1) + \frac{\zeta'(2m+1)}{\zeta(2m+1)} - \log(2\pi x), \end{aligned}$$

while (F.5) gives

$$(G.2) \quad \frac{1}{(2m)! \zeta'(-2m)} = \frac{2 (-1)^m (2\pi)^{2m}}{[(2m)!]^2 \zeta(2m+1)}.$$

Multiplying (G.1) by (G.2) and by  $x^{2m}$ , and using  $\psi(2m+1) = \Gamma'(2m+1)/(2m)!$ , we obtain exactly (4.13). The two computations of the residue at the double pole, the sine-splitting computation of Section 4 and the direct Laurent computation above, therefore agree identically.

#### APPENDIX H. THE GAMMA LOWER BOUND FROM THE WEIERSTRASS PRODUCT

We supply every step of Lemma 5.1. The Weierstrass product ([17, Chapter XII]) is

$$(H.1) \quad \frac{1}{\Gamma(z)} = z e^{\gamma_e z} \prod_{n \geq 1} \left( 1 + \frac{z}{n} \right) e^{-z/n},$$

convergent locally uniformly. At  $z = \sigma + it$ ,  $\sigma > 0$ , taking squared moduli and using  $|e^{\gamma_e z}|^2 = e^{2\gamma_e \sigma}$ ,  $|e^{-z/n}|^2 = e^{-2\sigma/n}$ ,  $|z|^2 = \sigma^2 + t^2$  and

$$|n + z|^2 = (n + \sigma)^2 + t^2,$$

$$(H.2) \quad \frac{1}{|\Gamma(\sigma + it)|^2} = (\sigma^2 + t^2) e^{2\gamma_e \sigma} \prod_{n \geq 1} \frac{(n + \sigma)^2 + t^2}{n^2} e^{-2\sigma/n}.$$

Dividing (H.2) by its value at  $t = 0$ , every  $t$ -independent factor cancels, and

$$(H.3) \quad \frac{|\Gamma(\sigma + it)|^2}{\Gamma(\sigma)^2} = \frac{\sigma^2}{\sigma^2 + t^2} \prod_{n \geq 1} \frac{(n + \sigma)^2}{(n + \sigma)^2 + t^2} = \prod_{n \geq 0} \left(1 + \frac{t^2}{(n + \sigma)^2}\right)^{-1},$$

which is (5.12). Taking logarithms and comparing the sum with the integral of the positive decreasing function  $u \mapsto \log(1 + t^2/u^2)$  over  $u \in (\sigma, \infty)$ , term  $n$  being bounded by the integral over  $(\sigma + n - 1, \sigma + n)$  for  $n \geq 1$ ,

$$(H.4) \quad \sum_{n \geq 0} \log \left(1 + \frac{t^2}{(n + \sigma)^2}\right) \leq \log \left(1 + \frac{t^2}{\sigma^2}\right) + \int_0^\infty \log \left(1 + \frac{t^2}{u^2}\right) du.$$

In the integral substitute  $u = |t|v$  and integrate by parts:

$$(H.5) \quad \int_0^\infty \log(1 + v^{-2}) dv = \left[v \log(1 + v^{-2})\right]_0^\infty + \int_0^\infty \frac{2dv}{1 + v^2} = 0 + \pi,$$

the boundary term vanishing at both ends because

$$\begin{aligned} v \log(1 + v^{-2}) &\sim 2v \log \frac{1}{v} \longrightarrow 0 \quad (v \rightarrow 0^+), \\ v \log(1 + v^{-2}) &\sim \frac{1}{v} \longrightarrow 0 \quad (v \rightarrow \infty). \end{aligned}$$

Hence the integral in (H.4) equals  $\pi|t|$ , and exponentiating,

$$(H.6) \quad \frac{|\Gamma(\sigma + it)|^2}{\Gamma(\sigma)^2} \geq \left(1 + \frac{t^2}{\sigma^2}\right)^{-1} e^{-\pi|t|},$$

which is the square of (5.11).

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